79. The jet aircrafts fly very easily and smoothly in the lower stratosphere. What could be the appropriate explanation?

1. There are no clouds or water vapour in the lower stratosphere.

2. There are no vertical winds in the lower stratosphere

Which of the statements given above/is are correct in this context?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (c) : In the lower part of the stratosphere, jet, planes can fly very smoothly as it does not have seasonal incidents like thunderstorms, thunder clouds, lightning strikes, water vapour and dust. The stratosphere ranges from 18 to 32 km. The ideal condition of flying for aircraft and jet aircraft is found in this stratosphere.

80. What is Virtual Private Network

(a) It is a private computer network of an organization where the remote users can transmit encrypted information through the server of the organization .

(b) It is a computer network across a public internet that provides users access to their organization's network while maintaining the security of the information transmitted.

(c) It is a computer network in which users can access a shared pool of computing resources through a service provider.

(d) None of the statements (a), (b), and (c) given above is correct description of Virtual Private Network

Ans. (b): A secure way to connect to a private Local Area Network (LAN) located at an organization's remote location is a virtual private network (VPN) in which the internet or an unsecured public network is used to exchange private information by encryption technology.

ENVIRONMENT & ECOLOGY

81. A sandy and saline area is the natural habitat of an Indian animal species. The animal has no predators in that area but its existence is threatened due to the destruction of its habitat . What one of the following could be that animal ?

(a) Indian wild buffalo (b) Indian wild ass

(c) Indian wild boar (d) Indian gazelle

Ans. (b) Sandy and saline regions are the natural habitats of an Indian animal species; such animal is an Indian wild Ass. Rann of Kutch is its natural habitat. There are no predators of that animal in this area, but due to the destruction of the habitat, its existence is in danger.

82. The ''Red Data Books'' published by the International Union for Conservation of Nature and Natural Resources (IUCN) contain lists of –

1. Endemic plant and animal species present in the biodiversity hotspots.

2. Threatened plant and animal species.

3. Protected sites for conservation of nature and natural resources in various countries.

Select the correct answer using the codes given below :

(a) 1 and 3 (b) 2 only

(c) 2 and 3 (d) 3 only

Ans. (b): The list of threatened plant and animal species are included in the "Red Data Book" published by the International Union for Conservation of Nature and Natural Resources (IUCN).

83. Three of the following criteria have contributed to the recognition of Western Ghats-Sri Lanka and Indo-Burma regions as hotspots of biodiversity:

1. Species richness

2. Vegetation density

3. Endemism

4. Ethno- botanical importance

5. Threat perception

6. Adaption of flora and fauna to warm and humid condition

Which three of the above are correct criteria in this context?

(a) 1, 2 and 6

(b) 2, 4 and 6

(c) 1, 3 and 5

(d) 3, 4 and 6

Ans. (c): The Western Ghats-Sri Lanka and Indo-Burma regions have been recognized as biodiversity hotspots based on three parameters called species richness, endemism and threat perception.

84. Human activities in the recent past have caused the increased concentrations of carbon dioxide in the atmosphere, but a lot of it does not remain in the lower atmosphere because of –

1. Its escape into the outer stratosphere.

2. The photosynthesis by phytoplankton in the oceans.

3. The trapping of air in the polar ice caps.

Which of the statements given above is/are correct?

(a) 1 and 2 (b) 2 only

(c) 2 and 3 (d) 3 only

Ans. (b): The concentration of carbon dioxide in the atmosphere has increased in recent years due to globalization rapidly increasing population, deforestation and other human activities, but most of it does not remain in the lower part of the atmosphere. One of the main reasons is that phyto plankton performs photosynthesis in the oceans.

85. In the context of ecosystem productivity, marine upwelling zones are important as they increase the marine productivity by bringing the –

1. Decomposer microorganisms to the surface.

2. Nutrients to the surface.

3. Bottom-dwelling organisms to the surface.

Which of the statements given above is/are correct?

(a) 1 and 2 (b) 2 only

(c) 2 and 3 (d) 3 only

Ans. (b) : In the context of productivity, the upwelling area is important because they increase marine productivity by bringing nutrients to the surface. Under the sea flow, the hot and nutritious water present on the sea surface is transferred by the airflow through cold and nutrient- rich water.

86. If a tropical rain forest is removed, it does not regenerate quickly as compared to a tropical deciduous forest. This is because –

(a) The soil or rain forest is deficient in nutrients.

(b) Propagules of the trees in a rain forest have poor viability.

(c) The rain forest species are slow-growing.

(d) Exotic species invade the fertile soil of rain forest

Ans. (a) : If a tropical rain forest is removed, it does not re-developed as fast as a tropical deciduous forest; the main reason being the lack of nutrients in the soil of the rain forest.

87. The Himalayan Range is very rich in species diversity. Which one among the following is the most appropriate reason for this phenomenon?

(a) It has a high rainfall that supports luxuriant vegetative growth.

(b) It is a confluence of different bio-geographical zones.

(c) Exotic and invasive species have not been introduced in this region.

(d) It has less human interference.

Ans. (b): The Himalayan Mountains form the natural and political boundary of the Indian sub-continent. Due to the geographical location of the Himalayas, the rest of the Indian subcontinent has become a distinct personality from Asia. The Himalayan mountains are extremely rich in terms of species. The main reason for that is that it is a confluence of different bio-geographical regions.

88. A layer in the Earth's atmosphere called Ionosphere facilitates radio communication. Why?

1. The presence of ozone caused the reflection of radio waves to Earth.

2. Radio waves have a very long wavelength.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (d): A layer called the Ionosphere in the Earth's atmosphere facilitates radio communication because of the presence of electrically charged ions in the ionosphere.

89. There is a concern over the increase in harmful algal blooms in the seawaters of India. What could be the causative factors for this phenomenon?

1. Discharge of nutrients from the estuaries.

2. Run-off from the land during the monsoon.

3. Upwelling in the seas.

Select the correct answer from the codes given below :

(a) 1 only (b) 1 and 2 only

(c) 2 and 3 only (d) 1, 2 and 3

Ans. (d): Concern has been expressed over the increase in harmful algae bloom in India's seawater. The main reason for this is the discharge of nutrients from the estuaries, run-off from the land during the monsoon and upwelling in the seas.